

Credit Card Fraud Detection

Can label-free anomaly detection match supervised models — and stay robust when fraud shifts?

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Geometric Learning, Time-Variant Data Analysis, and Anomaly Detection

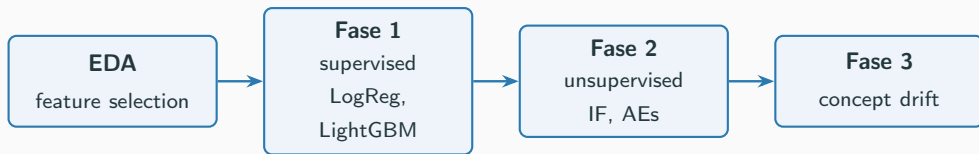
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The problem in one slide

Credit card fraud detection is structurally asymmetric: frauds are **rare**, labels are expensive, and attack patterns **evolve over time**.

Research question. Can *unsupervised* anomaly detection rival *supervised* learning — and stay robust when the fraud pattern shifts?



Exploratory Data Analysis

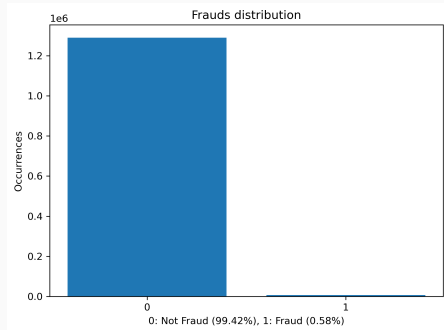
EDA approach

Screening method. Fraud vs legit are compared with the tool that fits each feature type; a feature is kept only if the classes are *stably separable*:

- **Categorical / temporal:** absolute + relative fraud count per category.
- **Continuous:** boxplots + empirical distributions.

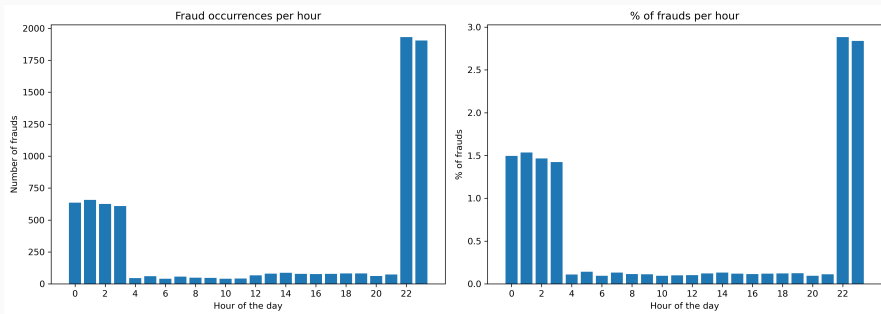
Clean dataset: 1 296 675 rows $\times$ 22 columns; 0 missing, 0 duplicates; only two columns need a datetime cast.

The target is severely imbalanced (0.58% fraud) — hence **PR-AUC** as the primary metric throughout, since accuracy and ROC-AUC would be misleading.



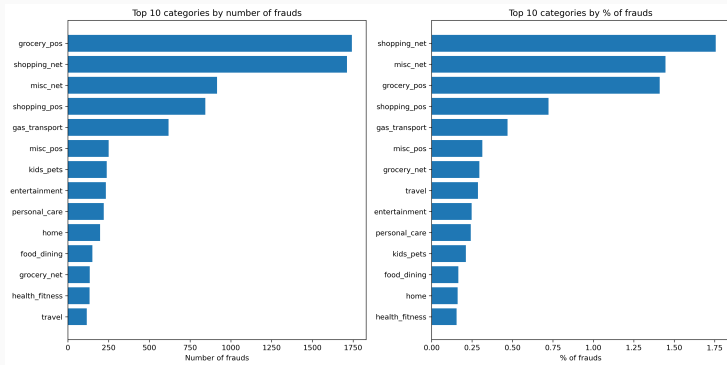
Class imbalance ($\sim 1:172$)

Signal 1 — Hour of day



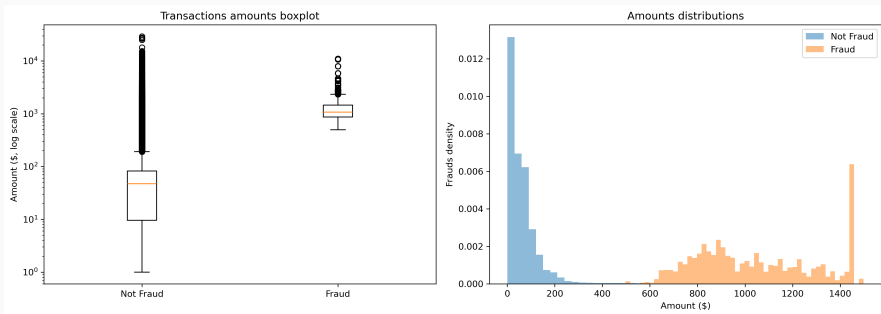
Fraud is strongly concentrated between **10 PM and 3 AM**, both in absolute count and as a share of hourly transactions. Day-of-week, day-of-month and month carry no consistent signal. The hour is encoded as **cyclical** (`hour_sin`, `hour_cos`) so that hour 23 and hour 0 sit next to each other.

Signal 2 — Category



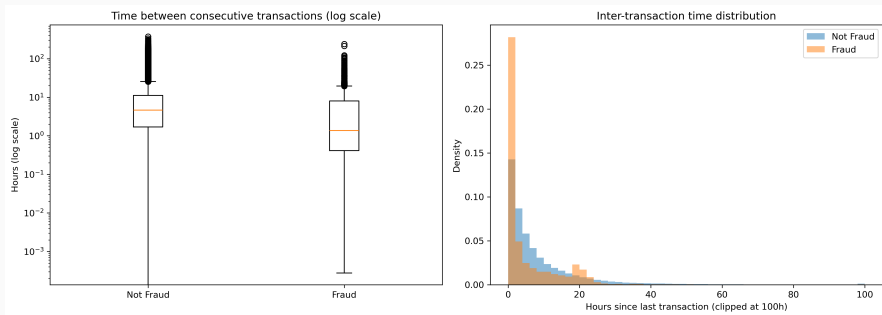
There are **14 transaction categories**. Counts are dominated by everyday categories, but the fraud rate concentrates in **online** channels (the `_net` suffix). Because the 14 levels interact differently with each model, they will require **different encodings** (one-hot, native categorical, embedding, or dropped — see the transformations slide).

Signal 3 — Amount



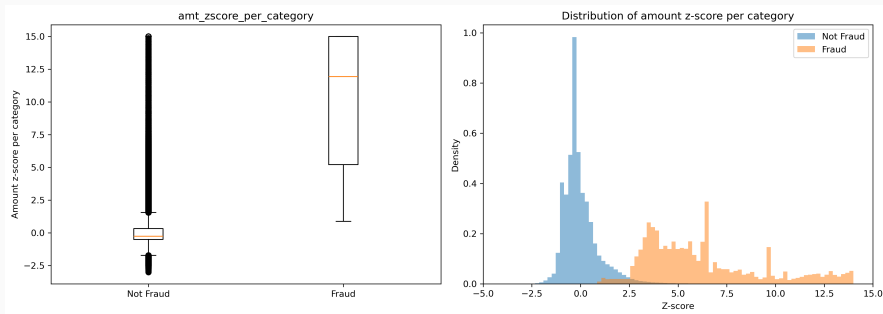
The **median** amount is **\$47** for legit vs **\$1 076** for fraud — a $\approx 23\times$ gap, i.e. strong distributional separation. The distribution is right-skewed, so amt gets a $\log_{10}$ transform for the scale-sensitive models (LogReg, AEs, IF).

Signal 4 — Inter-transaction time



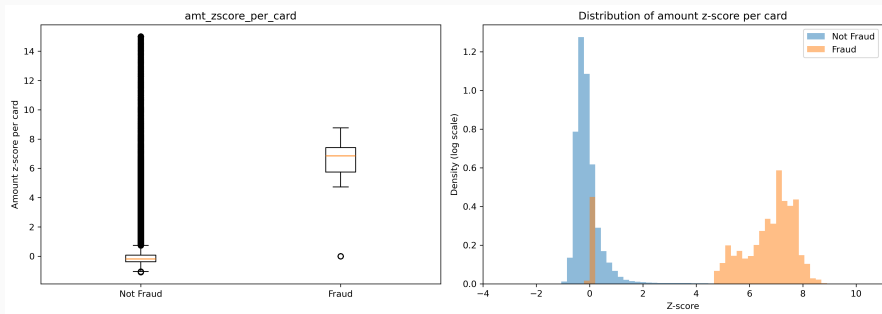
Fraud is **bursty**: the median gap between consecutive transactions on a card is **1.4 h** for fraud vs **4.6 h** for legit, a $\approx 3.3\times$ difference — suggesting that a stolen card is used intensively in a short window. Like `amt`, this feature is right-skewed, so it also gets a `log1p` transform for the scale-sensitive models.

Signal 5 — Amount z-score per category



Normalising the amount against the *category* baseline sharpens the `amt` signal: fraud sits at median $z \approx +12$, legit near zero. The baseline is a **fixed population statistic** — each category's mean/std computed *once* on the training set — so it is the same constant for every transaction in that category and needs no causal/expanding scheme. `amt_zscore_cat` captures **category-level** atypicality.

Signal 6 — Amount z-score per card



Normalising against the *card's owner* history captures **individual-level** atypicality: fraud median $z \approx +6.9$, legit near zero. Unlike the category baseline, this reference is card-specific, so a naive mean/std would let a fraudulent amount inflate its own baseline introducing leakage. The per-card statistics therefore use a **causal expanding window** ($1 \dots t-1$ only), frozen from training at test time.

EDA recap — features retained

Six core signals (plus category); the four engineered ones are derived only from the timestamp, the amount and the card/category keys.

Feature	EDA finding	Derived from
amt	fraud median $\sim \$1076$ is $23\times$ legit ($\sim \$47$)	raw
category	fraud concentrated in online _net channels	raw
hours_since_last_trans	bursty: median 1.4 h vs 4.6 h ($3.3\times$)	unix_time, cc_num
hour_sin, hour_cos	fraud peaks 10PM–3AM (cyclical)	trans_date_trans_time
amt_zscore_cat	atypical for the <i>category</i> ($z \approx +12$)	amt, category
amt_zscore_card	atypical for the <i>card</i> ($z \approx +6.9$, causal)	amt, cc_num

EDA recap — features dropped

The remaining columns showed no usable fraud signal and were dropped.

Feature(s)	Reason for dropping
merchant	~700 distinct values; high-rate merchants have very few transactions
job	~500 distinct values; high-rate jobs have very few transactions
dob, gender	demographic information shows no relationship with fraud
lat/long, merch_lat/long	home-to-merchant distance identical for fraud/legit
city_pop, state, city, zip	no geographic signal at any granularity
trans_num, cc_num, first, last	identifiers (cc_num used only to derive temporal features)
unix_time, trans_date_trans_time	replaced by engineered temporal features

EDA recap — transformations by model

Same feature set; each is encoded to match the model's inductive bias.

Feature	LogReg	LightGBM	Isolation Forest	Neural AEs
amt	log1p + scale	raw	log1p + scale	log1p + scale
hours_since_last_trans	log1p + scale	raw	log1p + scale	log1p + scale
hour	sin/cos + scale	raw	sin/cos + scale	sin/cos + scale
amt_zscore_cat	scale	raw	scale	scale
amt_zscore_card	scale	raw	scale	scale
category	one-hot	native cat.	— (dropped)	4-dim embedding

Fase 1 — Supervised models

Fase 1 — supervised setup

Goal. Measure how far supervision can go — the *ceiling* that the label-free models of Fase 2 must try to reach.

- **Data & split (chronological, shuffle=False):** train = first 80% of fraudTrain, validation = last 20%, test = fraudTest (the next 6 months). Time-ordered splits stop the model from learning on future transactions → no temporal leakage.
- **Metric: PR-AUC** (average precision), primary at 0.58% prevalence where ROC-AUC misleads; reported on validation *and* test.
- **Imbalance:** the minority class is up-weighted $\sim 170\times$ (`class_weight / scale_pos_weight`) so fraud is not ignored.
- **Models (full label access):** Logistic Regression (linear baseline) → LightGBM (non-linear champion).

Fase 1 — Logistic Regression (linear baseline)

Idea. Linear decision boundary on the encoded features; a deliberately simple reference for the non-linear models.

Key parameters.

- `class_weight='balanced'` reweights the minority class $\sim 170\times$ (else the model collapses to the majority class).
- L2 penalty (`C=1.0`); category via one-hot + `StandardScaler`.

Results — LogReg

	Val	Test
PR-AUC	0.7695	0.6606
ROC-AUC	0.9989	0.9987

Recall $\sim 100\%$ but precision only $\sim 36\%$: `balanced` maximises recall and floods false positives (val $\rightarrow$ test -14.2%). A simple **linear floor** — on test the Fase 2 autoencoders already beat it.

Fase 1 — LightGBM (non-linear champion)

Idea. Gradient-boosted trees on tabular data; captures *joint* non-linear interactions and splits categoricals natively.

Key parameters.

- `n_estimators=500`, `learning_rate=0.05`
- `scale_pos_weight≈170` (handles the imbalance)
- early stopping (patience 50); raw features — no scaling or encoding.

Results — LightGBM

	Val	Test
PR-AUC	0.9718	0.9420
ROC-AUC	0.9998	0.9986

Precision $\sim 68\%$ and recall $\sim 0.94\%$ on test — precise and complete, unlike LogReg. Smallest val $\rightarrow$ test drop (-3.1%): it fires only when several signals coincide (late-night *and* bursty *and* extreme card z-score).

The supervised ceiling.

Fase 1 — summary

Model	Val PR-AUC	Test PR-AUC	Val ROC	Test ROC
Logistic Regression	0.7695	0.6606	0.9989	0.9987
LightGBM	0.9718	0.9420	0.9998	0.9986

- **LightGBM sets the supervised ceiling** at PR-AUC 0.97 val / 0.94 test.
- LogReg (0.77 val / 0.66 test) is a deliberately simple **linear reference**.
- **The question for Fase 2:** can a model reach this *without any fraud label*?

Fase 2 — Unsupervised models

Fase 2 — unsupervised setup

Goal. Detect fraud with *no labels* — learn what *normal* looks like and flag deviations. Can a label-free model reach the Fase 1 ceiling?

- **Training (normal only):** models see *only* legitimate transactions ($\sim 1\text{M}$ pool; neural nets on a 200K subsample) — fraud is never used to fit.
- **Scoring & threshold:** anomaly score (higher = more anomalous); operating threshold = **99.5th percentile** of training scores. Same PR-AUC metric and chronological split as Fase 1.
- **Sequence framing:** each transaction is the last step of a **10-step card history** (96.4% of cards have ≥ 10), giving the sequential models temporal context.
- **Models (increasing context):** Isolation Forest $\rightarrow$ MixedAE $\rightarrow$ LSTM-AE $\rightarrow$ GRU-AE.

2.1 Isolation Forest

Idea. Ensemble of random trees; anomalies are isolated in *fewer* splits $\Rightarrow$ shorter path length $\Rightarrow$ higher score. No distributional assumption.

Key parameters.

- `n_estimators=100` (more trees give only diminishing returns).
- Operates on the 6 numeric features only — category excluded.

Results

	Val	Test
PR-AUC	0.6114	0.5206
ROC-AUC	0.9974	0.9974

Test PR-AUC **0.52**, precision 0.43 / recall 0.85 (−15% val $\rightarrow$ test): the numeric-only boundary is too diffuse for a tight threshold. **The unsupervised floor.**

2.2 MixedAE — per-transaction autoencoder

Idea. Compress & reconstruct a single transaction; large reconstruction error = far from the normal manifold.

Key parameters.

- Input: 6 numeric + category embedding (dim 4) = **10-dim**.
- Encoder $10 \rightarrow 64 \rightarrow 32 \rightarrow 4$ (latent), decoder $4 \rightarrow 32 \rightarrow 64 \rightarrow 6$; ReLU.
- MSE on the 6 numeric features; Adam $1r=1e-3$, 30 epochs, 200K-sample.

Results

	Val	Test
PR-AUC	0.6608	0.6768
ROC-AUC	0.9727	0.9814

Test PR-AUC **0.68**: big non-linear jump over IF (+0.16), *already above* supervised LogReg (0.66); precision 0.55 / recall 0.68. Still blind to each card's history — it flags only *globally* unusual amounts.

2.3 LSTM-AE — sequential autoencoder

Idea. An LSTM reads the 9 prior transactions of the card and predicts the numerical features of the current one. Error is high when the transaction breaks the card's *established* pattern.

Key parameters.

- LSTM(input=10, hidden=64) + MLP head; MSE on the numeric features.
- Adam lr=1e-3, **80 epochs** (150 over-trains — it learns to reconstruct anomalies too).

Results

	Val	Test
PR-AUC	0.7783	0.7074
ROC-AUC	0.9976	0.9963

Test PR-AUC **0.71** (best label-free), recall 0.91 / precision 0.35 (−9% val→test). Temporal context adds little here (+0.03 over MixedAE on test); its real payoff is **robustness under drift**.

2.4 GRU-AE — cell-mechanism ablation

Idea. An ablation of LSTM-AE: swap the recurrent cell, keep everything else, to test whether the gate design matters.

Key parameters.

- GRU cell in place of LSTM: $\sim 25\%$ fewer parameters (merged forget/input gates).
- Identical training: 10-step window, hidden 64, Adam $1r=1e-3$, 80 epochs.

Results

	Val	Test
PR-AUC	0.7777	0.7032
ROC-AUC	0.9978	0.9968

Test PR-AUC **0.70**, precision 0.34 / recall 0.89 (-10% val $\rightarrow$ test) — within noise of LSTM-AE: swapping the recurrent cell changes nothing.

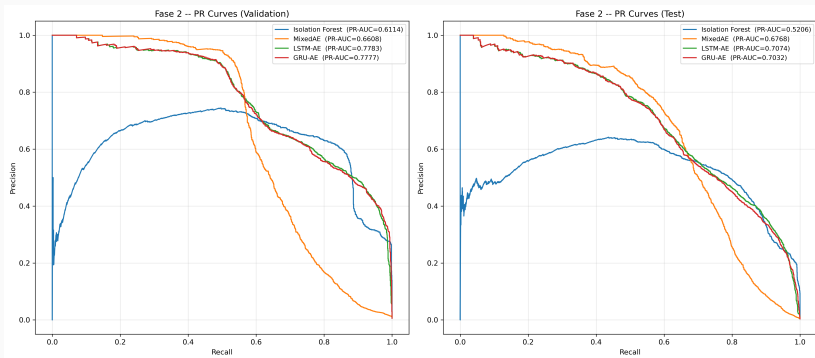
Fase 2 — summary

Model	Val PR-AUC	Test PR-AUC	Val ROC	Test ROC
Isolation Forest	0.6114	0.5206	0.9974	0.9974
MixedAE	0.6608	0.6768	0.9727	0.9814
LSTM-AE	0.7783	0.7074	0.9976	0.9963
GRU-AE	0.7777	0.7032	0.9978	0.9968

Added ingredient	Δ Val	Δ Test
IF $\rightarrow$ MixedAE	+0.049	+0.156
MixedAE $\rightarrow$ LSTM-AE	+0.118	+0.031
LSTM-AE $\rightarrow$ GRU-AE	-0.001	-0.004

Key finding (test): all three autoencoders beat supervised LogReg (0.66) — already at **MixedAE** (0.68). The big in-distribution gain is the non-linear reconstruction; temporal context adds little here (validation overstates it) and the cell type nothing. Sequence pays off under *drift*.

Fase 2 — PR curves (validation vs test)



The two sequential models (LSTM/GRU-AE) trace a high **cliff** (precision near 1 up to recall ≈ 0.55); MixedAE sits clearly below, and Isolation Forest is weakest. The **ordering is identical on validation (left) and test (right)** — the ranking holds out-of-time.

Fase 1 vs Fase 2 — where we stand

Model	Type	Val PR-AUC	Test PR-AUC
LightGBM	Supervised	0.9718	0.9420
LSTM-AE	Unsupervised	0.7783	0.7074
GRU-AE	Unsupervised	0.7777	0.7032
MixedAE	Unsupervised	0.6608	0.6768
Logistic Regression	Supervised	0.7695	0.6606
Isolation Forest	Unsupervised	0.6114	0.5206

- **In distribution, labels still win big:** LightGBM (0.94 test) is far above the best label-free model (LSTM-AE, 0.71).
- **But labels aren't needed to beat the linear baseline:** on test *all three* autoencoders top LogReg (0.66) — already MixedAE (0.68).
- **Among label-free models:** LSTM $\approx$ GRU > MixedAE > IF, but the sequential edge over MixedAE is small here (+0.03 test).
- In-distribution only $\rightarrow$ **Fase 3** tests drift, where that small edge becomes decisive.

Fase 3 — Concept drift

Fase 3 — simulating concept drift

Setup. On the held-out test set, every fraud amount is overwritten with exactly $\mu_{\text{card}} + 6\sigma_{\text{card}}$ (the card's training mean + 6 std). This makes the amount signal **weaker and more uniform**; all other features unchanged. **No retraining** — all six models scored as-is.

Hypothesis. Supervised models memorise the *specific* fraud signature and should degrade; models that represent *normality* (geometry or card history) should be more robust.

Baseline = original fraudTest PR-AUC; we measure relative loss after the 6σ shift.

Fase 3 — results (6σ drift)

Model	Type	Original	6σ drift	Rel. loss
LightGBM	Supervised	0.9420	0.0030	−99.7%
Logistic Regression	Supervised	0.6606	0.4089	−38.1%
MixedAE	Unsupervised	0.6768	0.4165	−38.5%
LSTM-AE	Unsupervised	0.7074	0.5268	−25.5%
GRU-AE	Unsupervised	0.7032	0.5416	−23.0%
Isolation Forest	Unsupervised	0.5206	0.4113	−21.0%

- **LightGBM collapses** (−99.7%): it memorised the exact amount threshold; at 6σ the signal *inverts* and it misranks fraud as legit.
- **LogReg** $\approx$ **MixedAE** ($\approx -38\%$): both lean on *absolute* amount magnitude — being unsupervised does *not* help here.
- **Sequential & IF most robust**: *relative* card history and geometric isolation survive an amount shift.

Fase 3 — what survives, and one caveat

The finding. All the autoencoders learn a model of normality — but *what* they call normal differs. MixedAE learns a **global** baseline: a transaction is anomalous if its amount is unusual across the whole population. LSTM/GRU-AE learn a **card-relative** baseline: a transaction is anomalous if it breaks *that card's own* spending pattern. When the drift uniformises absolute amounts, the global baseline loses its signal; the card-relative one, anchored to each card's history, survives.

Why it matters. The *best* model in distribution (LightGBM) is the *most fragile* out of it — scoring high on today's fraud says little about tomorrow's.

One honest caveat. The drift moved only the amount and left the timing signal intact, so part of the sequential models' edge may come from that untouched feature. Repeating the test on the timing would settle it — the amount shift is simply the most realistic place to start.

1. **Labels help, but aren't the whole story.** LightGBM wins in distribution (PR-AUC 0.94 test), yet on test the label-free autoencoders already beat the supervised *linear* baseline (LogReg 0.66).
2. **Robustness comes from the *scope* of the learned normality.** MixedAE is unsupervised yet breaks under drift like LogReg, because both rely on a *global* distribution of amounts. LSTM/GRU-AE compare each transaction against that card's own history — a reference the drift cannot erase.
3. **Practical takeaway.** Where fraud keeps shifting, models that anchor to *individual* behavioural patterns (sequential AEs, IF) are the defensible choice — not because they skip labels, but because of the normality they model.

Thank you.